JEE Mathematics: Quick Review with Grading and Practice Map - Continuity & Differentiability (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (51 questions; a question that appears in two of these is counted once): High = 5 or more questions, Medium = 1 to 4 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Definition of Continuity

Question 01

What is the graphical definition of continuity?

Model Answer

f is continuous at $x = a$ if its graph can be drawn through $x = a$ without lifting the pen (no break, no hole, no jump).

Ex: $y = \sin x,\ \cos x,\ |x|$ are continuous everywhere; $y = [x]$ and $y = \{x\}$ break at every $x \in I$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 02

What is the mathematical definition of continuity?

Model Answer

$$\lim_{h\to 0} f(a-h) = \lim_{h\to 0} f(a+h) = f(a), \quad h > 0$$

i.e. LHL = RHL = $f(a)$ (all three finite). To find a constant, equate the one-sided limits with $f(a)$.

Ex: $f = x + 1\ (x \le 1),\ 6 - kx^2\ (x > 1)$ continuous at 1: $2 = 6 - k \Rightarrow k = 4$.

Grade - High

Practice map, main concept: Sheet: J-I 1; J-II 1-4, 6-9, 12, 38; J-M 2, 5, 12, 16, 20-22, 24, 26; J-A 1; C-I 2, 4, 8; C-II 2, 5-6, 8, 15, 22 | DPP: 1(1-3, 5-10, 12-15, 17-19); 2(1-2, 4, 10, 12-17) | Question Bank: 118-119, 126, 129

Also used in: Sheet: J-I 2-6, 10, 12; J-II 5, 13, 16, 22, 24, 29-30, 32-33, 42, 44, 50, 59; J-M 9, 19; J-A 8; C-I 1, 3, 5-6, 17-18; C-II 3-4, 7 | DPP: 1(4); 2(5-7, 11); 3(1, 3, 6, 8, 10-12); 4(8, 11) | Question Bank: 113

Question 03

What to keep in mind while checking continuity if one side of the point is not in the domain?

Model Answer

Only the side inside the domain is checked: at the end point a (domain on the right) f is continuous if RHL = $f(a)$.

Ex: $\sqrt{x}$ is continuous at 0 (RHL = $0 = f(0)$); $\sin^{-1} x$ is continuous at 1 (LHL = $\frac{\pi}{2} = f(1)$).

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-I 3; J-M 11, 23 | Question Bank: 117

Important Points related to Continuity

Question 04

What is the relation between existence of limit and continuity at a point?

Model Answer

Continuous $\Rightarrow$ limit exists; limit exists $\not\Rightarrow$ continuous. Continuity is discussed only at points of the domain (a must be in the domain).

Ex: $[x] + [-x]$ at 0: limit $= -1$ but $f(0) = 0$, so not continuous. $\frac{1}{x}$ is continuous (in its domain).

Grade - **Low**

Practice map, main concept: none

Also used in: DPP: 2(8)

Question 05

What is the condition for a function to be left continuous at $x = a$?

Model Answer

$$\lim_{x \to a^-} f(x) = f(a)$$

Ex: $[x]$ at $x = 1$: LHL $= 0 \neq f(1) = 1$, so $[x]$ is not left continuous at 1.

Grade - **Low**

Practice map, main concept: none

Also used in: none

Question 06

What is the condition for a function to be right continuous at $x = a$?

Model Answer

$$\lim_{x \to a^+} f(x) = f(a)$$

Ex: $[x]$ at $x = 1$: RHL $= 1 = f(1)$, so $[x]$ is right continuous at every integer.

Grade - **Low**

Practice map, main concept: none

Also used in: none

Question 07

Comment on continuity of $f + g$, $f - g$, $f \cdot g$, f/g if both f and g are continuous at a point.

Model Answer

All four are continuous (f/g needs $g(a) \neq 0$). Polynomial, trigonometric, exponential and logarithmic functions are continuous in their domains.

Ex: $\dfrac{x^2 + 1}{x - 2}$ is continuous for all $x \neq 2$.

Grade - **Low**

Practice map, main concept: none

Also used in: Sheet: J-II 48; C-II 1

Question 08

Comment on continuity of $f+g$, $f-g$, $f\cdot g$, f/g if both f and g are discontinuous at a point.

Model Answer

Nothing definite can be said about any of them; check directly.

Ex: $[x]$, $\{x\}$ are discontinuous at 1, but $[x]+\{x\}=x$ and $[x]\{x\}$ are continuous at 1; $[x]\{x\}$ is discontinuous at 0 (LHL $=-1$, RHL $=0$).

Grade - Low

Practice map, main concept: Sheet: J-I 5

Also used in: none

Question 09

Comment on continuity of $f+g$, $f-g$, $f\cdot g$, f/g if f is continuous and g is discontinuous at a point.

Model Answer

$f\pm g$ is surely discontinuous; nothing definite about $f\cdot g$ and f/g.

Ex: $x[x]$ is continuous at $x=0$ (both limits 0) but discontinuous at $x=1$ (LHL $=0$, RHL $=1$).

Grade - Low

Practice map, main concept: Sheet: J-I 4

Also used in: DPP: 2(7)

Continuity in an Interval

Question 10

What is the condition for a function to be continuous in (a,b)?

Model Answer

f is continuous at each and every point of (a,b).

Ex: $\frac{1}{x}$ is continuous in $(0,1)$ although it is unbounded there.

Grade - Medium

Practice map, main concept: Sheet: J-A 14

Also used in: none

Question 11

What is the condition for a function to be continuous in $[a,b]$?

Model Answer

(i) continuous in (a,b), (ii) right continuous at a, (iii) left continuous at b.

Ex: $[x]$ is not continuous in $[0,1]$ (not left continuous at 1); $\sqrt{1-x^2}$ is continuous in $[-1,1]$.

Grade - Low

Practice map, main concept: Sheet: J-I 14

Also used in: Sheet: J-II 10

Question 12

What are the checkpoints for continuity in an interval?

Model Answer

(i) points where the definition changes, (ii) for $[f(x)]$, $\{f(x)\}$: points where $f(x)$ is an integer, (iii) for $\operatorname{sgn} f(x)$: points where $f(x) = 0$, (iv) points where a denominator is 0 or the domain breaks.

Ex: $\operatorname{sgn}(x-2) + [x - \frac{1}{2}] + [\ln x]$ on $[1, 3]$ is discontinuous at $x = \frac{3}{2},\ 2,\ \frac{5}{2},\ e$.

Grade - High

Practice map, main concept: Sheet: J-I 2, 12; J-II 10-11, 14-15, 39, 48, 50, 54-55; J-M 11, 23, 25; J-A 8, 15; C-II 21 | DPP: 1(4); 2(3, 6-8) | Question Bank: 117, 125, 128

Also used in: Sheet: J-I 14; J-II 2, 27, 31, 37, 45, 59-61; J-M 5, 7, 10; J-A 7; C-I 15-16, 22; C-II 10, 14 | DPP: 2(1); 3(2, 7); 4(1, 9-10, 13) | Question Bank: 127

Question 13

How to check continuity of a function defined separately for rational and irrational numbers?

Model Answer

$f = g(x)$ for $x \in Q$, $h(x)$ for $x \notin Q$ (g, h continuous): f is continuous only at the points where $g(x) = h(x)$; discontinuous at all other points.

Ex: $f = x^2 + 6$ $(x \in Q)$, $5x$ $(x \notin Q)$: $x^2 - 5x + 6 = 0$, continuous only at $x = 2, 3$.

Grade - Low

Practice map, main concept: Sheet: C-I 9, 12 | DPP: 1(11)

Also used in: Sheet: C-I 1

Question 14

How to check continuity of $[f(x)]$ if the graph of f can be drawn?

Model Answer

Draw $y = f(x)$ and the lines $y = n$ $(n \in I)$. $[f]$ is discontinuous where the graph crosses an integer line, or touches it from below (local max at an integer value); touching from above (local min) keeps $[f]$ continuous.

Ex: $[\sin x]$ in $[0, \pi]$ is discontinuous only at $x = \frac{\pi}{2}$.

Grade - Medium

Practice map, main concept: Sheet: C-I 7

Also used in: Sheet: J-II 10, 28; J-M 8, 10; J-A 7, 9; C-II 7, 10, 21, 23 | DPP: 2(3); 4(9)

Question 15

How to find continuity of functions with built-in limit?

Model Answer

First evaluate the limit separately for $|x| < 1$, $|x| = 1$, $|x| > 1$ ($x^{2n} \to 0,\ 1,\ \infty$), write f piecewise, then check continuity at $x = \pm 1$.

Ex: $f = \lim_{n\to\infty} \dfrac{x^{2n-1} + ax^2 + bx}{x^{2n} + 1}$: $ax^2 + bx$ $(|x| < 1)$, $\frac{1}{x}$ $(|x| > 1)$; continuous on R iff $a = 0,\ b = 1$.

Grade - Low

Practice map, main concept: Sheet: J-II 13; C-I 1, 5, 17; C-II 1, 3-4, 7

Also used in: none

Types of Discontinuity

Question 16

What is the condition for removable type of discontinuity?

Model Answer

$\lim_{x\to a} f(x)$ exists (finite) but $\neq f(a)$ or $f(a)$ is not defined. Redefine $f(a) = \lim_{x\to a} f(x)$ to make f continuous.

Ex: $\dfrac{x^2-4}{x-2}$ at $x = 2$: limit 4; define $f(2) = 4$.

Grade - **Low**

Practice map, main concept: Sheet: J-I 6; J-II 46; C-I 18

Also used in: Sheet: J-II 4, 49, 60 | DPP: 1(9); 2(2, 10, 17)

Question 17

What is the condition for non-removable type of discontinuity?

Model Answer

$\lim_{x\to a} f(x)$ does not exist (LHL $\neq$ RHL, or a one-sided limit is infinite or oscillates); no redefinition of $f(a)$ can help.

Ex: $[x]$ at $x = 1$ (LHL $= 0$, RHL $= 1$).

Grade - **Low**

Practice map, main concept: Sheet: J-II 5, 17, 49; C-I 3

Also used in: Sheet: J-II 46; C-II 19

Question 18

What are the different types of removable discontinuity and the condition for each?

Model Answer

Missing point: limit exists, $f(a)$ not defined. Isolated point: limit exists, $f(a)$ defined but $\neq$ limit.

Ex: $\frac{\sin x}{x}$ at 0: missing point. $[x] + [-x]$ at 0: isolated point (limit -1, $f(0) = 0$).

Grade - **Low**

Practice map, main concept: none

Also used in: Sheet: J-II 46; C-I 18

Question 19

What are the different types of non-removable discontinuity and the condition for each?

Model Answer

Finite (jump): both one-sided limits finite and unequal. Infinite: at least one one-sided limit is $\pm\infty$. Oscillatory: the limit oscillates between two finite values.

Ex: $\tan^{-1}\frac{1}{x}$ at 0: $\pm\frac{\pi}{2}$, finite; $\frac{1}{x^2}$ at 0: infinite; $\sin\frac{1}{x}$ at 0: oscillatory.

Grade - **Low**

Practice map, main concept: Sheet: C-II 19 | DPP: 2(11)

Also used in: Sheet: J-II 17, 49, 53 | DPP: 2(9, 13-15)

Question 20
What is the jump of discontinuity in finite type discontinuity?

Model Answer

$$\text{Jump} = |\text{RHL} - \text{LHL}|$$

Ex: $\frac{|\sin x|}{x}$ at 0: ± 1, jump 2; $\frac{[x]}{x}$ at 2: 1 and $\frac{1}{2}$, jump $\frac{1}{2}$.

Grade - Low
Practice map, main concept: DPP: 2(9)
Also used in: Sheet: J-II 53; C-II 19 | DPP: 2(11)

Question 21
What are the checkpoints for continuity of $f(g(x))$ if the checkpoints for f and g are known?

Model Answer
If f is discontinuous at $x = a, b$ and g at $x = c, d$: check $f(g(x))$ at $x = c, d$ and at all x with $g(x) = a$ or $g(x) = b$. (For $f \pm g$: a, b, c, d are all discontinuities; for fg, f/g they are checkpoints.)
Ex: $f(x) = \frac{1}{1-x}$: $f(f(x)) = \frac{x-1}{x}$ is discontinuous at $x = 1$ and at $x = 0$ (where $f(x) = 1$).

Grade - Medium
Practice map, main concept: Sheet: J-I 3; J-II 61; C-II 11, 14 | DPP: 2(5, 18)
Also used in: Sheet: J-II 48, 56; J-M 6; J-A 13; C-I 4 | DPP: 1(11)

Question 22
State the extreme value theorem.

Model Answer
If f is continuous on $[a, b]$, then f attains a least value m and a greatest value M on $[a, b]$ (so f is bounded and its range is $[m, M]$).
Ex: x^2 on $[-1, 2]$: $m = 0$, $M = 4$. On $(0, 1]$, $\frac{1}{x}$ has no greatest value (interval not closed).

Grade - Low
Practice map, main concept: Sheet: C-I 19
Also used in: Sheet: C-I 1

Intermediate Value Theorem (IMVT)

Question 23
Write the statement of IMVT.

Model Answer
If f is continuous on $[a, b]$ and $f(a) \neq f(b)$, then for every c between $f(a)$ and $f(b)$ there is at least one $x_0 \in (a, b)$ with $f(x_0) = c$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: C-I 14

Question 24
Write the alternate statement of IMVT.

Model Answer
If f is continuous on $[a,b]$ and $f(a)f(b)<0$, then $f(x)=0$ has at least one root in (a,b) (there may be more). The condition is sufficient, not necessary.
Ex: x^3+x-1: $f(0)=-1$, $f(1)=1$, so a root lies in $(0,1)$.

Grade - **Low**
Practice map, main concept: Sheet: J-II 18, 47; C-I 14
Also used in: Sheet: C-I 19 | DPP: 4(5)

Question 25
How to determine whether a question is of IMVT?

Model Answer
Only continuity and end values are given and the question asks to show that an equation has at least one solution in an interval, or that some c exists with $f(c)=\ldots$.

Grade - **Low**
Practice map, main concept: none
Also used in: Sheet: C-I 14

Question 26
How to apply IMVT in questions?

Model Answer
Bring all terms to one side: $g(x)=\text{LHS}-\text{RHS}$; check g is continuous on $[a,b]$; show $g(a)g(b)\le 0$ (for $f(c)=c$ take $g=f(x)-x$; for $f(c)=f(c+k)$ take $g=f(x+k)-f(x)$).
Ex: $(x-1)(x-3)=(x-2)(x-4)$: $g(2)=-1$, $g(4)=3$, so a root lies in $(2,4)$.

Grade - **Low**
Practice map, main concept: Sheet: J-II 19-20 | DPP: 1(16); 4(5)
Also used in: Sheet: J-II 18, 47

Definition of Differentiability

Question 27
What is the relation between the tangent at a point and derivability of the function at that point?

Model Answer
f is derivable at $x=a$ iff a unique tangent of finite slope exists there. A vertical tangent ($x^{1/3}$ at 0) or no unique tangent ($|x|$ at 0) means not derivable.

Grade - **Low**
Practice map, main concept: none
Also used in: Sheet: C-I 10

Question 28
Write the expressions of LHD and RHD.

Model Answer

$$\text{RHD}=f'(a^+)=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h},\qquad \text{LHD}=f'(a^-)=\lim_{h\to 0}\frac{f(a-h)-f(a)}{-h},\quad h>0$$

Each must exist and be finite.
Ex: $|x|$ at 0: LHD $=-1$, RHD $=1$.

Grade - Medium

Practice map, main concept: Sheet: J-I 9; J-II 23, 26, 30; C-I 6

Also used in: Sheet: J-I 8; J-II 22, 58; J-A 2, 9, 15, 17; C-II 12 | DPP: 3(10); 4(7)

Question 29

Write all the different ways in which LHD and RHD can be written.

Model Answer

$$\text{RHD} = \lim_{x\to a^+} \frac{f(x)-f(a)}{x-a} = \lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h}, \qquad \text{LHD} = \lim_{x\to a^-} \frac{f(x)-f(a)}{x-a} = \lim_{h\to 0^+} \frac{f(a)-f(a-h)}{h}$$

If f is derivable at a: $\lim_{h\to 0} \dfrac{f(a+ph)-f(a+qh)}{h} = (p-q)f'(a)$.

Ex: $f'(2) = 3 \Rightarrow \lim_{h\to 0} \dfrac{f(2+h)-f(2-h)}{h} = 6$.

Grade - High

Practice map, main concept: Sheet: J-I 15, 22-23; J-II 43, 51, 58; J-A 10; C-I 10; C-II 9, 13 | DPP: 3(4); 4(6, 8)

Also used in: Sheet: J-II 32-33, 35-36, 44; J-M 18; J-A 5, 11-12; C-I 11 | Question Bank: 130

Question 30

What is the condition for a function to be derivable at $x = a$?

Model Answer

$$\text{LHD} = \text{RHD} = \text{finite} = f'(a)$$

For unknown constants: first make f continuous at a, then equate LHD and RHD.

Ex: $f = ax + b$ $(x \le -1)$, $ax^5 + x + 2b$ $(x > -1)$ derivable: $b = 1$, $a = 5a + 1 \Rightarrow a = -\frac{1}{4}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 10; J-II 24, 29, 32-33, 44, 59; J-M 9, 19 | DPP: 3(3, 6, 8, 10, 12); 4(7, 11)

Also used in: Sheet: J-II 21, 52, 56, 60; C-I 8, 22

Question 31

What to keep in mind while checking derivability if one side of the point is not in the domain?

Model Answer

Only the one-sided derivative from the domain side is needed: at a left end point check RHD only (finite $\Rightarrow$ derivable there).

Ex: $\sqrt{x}$ at 0: RHD $= \lim \frac{1}{\sqrt{h}} = \infty$, not derivable; $x^{3/2}$ at 0: RHD $= 0$, derivable.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 32

What is the relation between continuity and derivability at a point?

Model Answer

Derivable $\Rightarrow$ continuous; continuous $\not\Rightarrow$ derivable; discontinuous $\Rightarrow$ not derivable; not derivable $\not\Rightarrow$ discontinuous.

Ex: $|x|$ is continuous but not derivable at 0.

Grade - Medium

Practice map, main concept: Sheet: J-II 31, 60

Also used in: Sheet: J-I 23; J-II 26, 37, 40, 45, 54; J-M 8; J-A 3 | DPP: 3(3)

Question 33
What is the condition for a function to be left derivable at $x = a$?

Model Answer
LHD at $x = a$ exists and is a finite number.
Ex: $[x]$ at 1: LHD $= \lim \frac{0-1}{-h} = \infty$, not left derivable.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 34
What is the condition for a function to be right derivable at $x = a$?

Model Answer
RHD at $x = a$ exists and is a finite number.
Ex: $[x]$ at 1: RHD $= \lim \frac{1-1}{h} = 0$, right derivable.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 35
Comment on derivability of $f + g$, $f - g$, $f \cdot g$, f/g if both f and g are derivable at a point.

Model Answer
All four are derivable (f/g needs $g(a) \neq 0$). Polynomial, trigonometric, exponential and logarithmic functions are derivable in their domains.

Grade - Low
Practice map, main concept: none
Also used in: DPP: 4(3)

Question 36
Comment on derivability of $f + g$, $f - g$, $f \cdot g$, f/g if both f and g are non-derivable at a point.

Model Answer
Nothing definite; check directly.
Ex: at 0, $|x| + (-|x|) = 0$ is derivable but $|x| + |x|$ is not.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 37
Comment on derivability of $f + g$, $f - g$, $f \cdot g$, f/g if f is derivable and g is non-derivable at a point.

Model Answer
$f \pm g$ is surely non-derivable; nothing definite about $f \cdot g$, f/g.
Ex: $x|x|$ is derivable at 0, but $(x + 1)|x|$ is not (LHD $= -1$, RHD $= 1$).

Grade - Medium
Practice map, main concept: Sheet: J-A 11
Also used in: none

Question 38
What is the condition for a function to be derivable in (a, b)?

Model Answer
f is derivable at each and every point of (a, b).

Grade - Medium
Practice map, main concept: Sheet: C-II 16
Also used in: Sheet: J-II 55; J-A 14

Question 39
What is the condition for a function to be derivable in $[a, b]$?

Model Answer
(i) derivable in (a, b), (ii) right derivable at a, (iii) left derivable at b.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 40
What are the checkpoints for derivability in an interval?

Model Answer
(i) all checkpoints of continuity (discontinuous there $\Rightarrow$ not derivable), (ii) for $|f(x)|$: points where $f(x) = 0$, (iii) points where a piece has a vertical tangent, e.g. $(x - a)^{1/3}$.
Ex: $|x - 1| + |x - 2| + |x - 3|$ is not derivable at $x = 1, 2, 3$.

Grade - High
Practice map, main concept: Sheet: J-I 11, 13; J-II 27-28, 37, 40, 57; J-M 7-8, 10; J-A 7; C-I 16, 23 | DPP: 3(2, 7, 14); 4(9) | Question Bank: 127
Also used in: Sheet: J-I 17; J-M 17; J-A 4, 6, 13, 15; C-I 7, 13, 21; C-II 11, 21 | DPP: 4(10) | Question Bank: 123

Question 41
How to check derivability of a function defined separately for rational and irrational numbers?

Model Answer
$f = g(x)$ $(x \in Q)$, $h(x)$ $(x \notin Q)$ can be derivable only at a point a where $g(a) = h(a)$ and $g'(a) = h'(a)$; check there by the definition.
Ex: x^2 (Q), 0 (Q') is derivable at 0 $(f'(0) = 0)$; x (Q), 0 (Q') is continuous but not derivable at 0.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: C-I 12

Question 42
Comment on continuity and differentiability of $x^p \sin \frac{1}{x}$, $x^p \cos \frac{1}{x}$ at $x = 0$ (value 0 at $x = 0$).

Model Answer
Continuous at 0 iff $p > 0$; derivable at 0 iff $p > 1$ $(f'(0) = 0)$; f' continuous at 0 iff $p > 2$.
Ex: $x^2 \sin \frac{1}{x}$ is derivable at 0 with $f'(0) = 0$, but $f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x}$ is not continuous at 0.

Grade - High
Practice map, main concept: Sheet: J-I 8; J-II 21; J-M 1; J-A 9, 16-17; C-I 20; C-II 12
Also used in: Sheet: J-A 2

Concept of LHD and RHD

Question 43

What can you say about continuity from the values of LHD and RHD?

Model Answer

LHD = RHD = finite: derivable and continuous. Both finite but unequal: continuous, not derivable. At least one infinite: not derivable, continuity cannot be decided.

Ex: $x^{1/3}$ at 0 (both ∞) is continuous; $\operatorname{sgn} x$ at 0 (both ∞) is discontinuous.

Grade - Low

Practice map, main concept: DPP: 3(5)

Also used in: none

Question 44

What is the condition for a sharp corner to exist in the graph of a function?

Model Answer

f continuous at a and LHD $\neq$ RHD (no unique tangent). Sharp corner $\Rightarrow$ not derivable, but not derivable $\not\Rightarrow$ sharp corner (vertical tangent of $x^{1/3}$).

Ex: $|x|$ and $|x|^{1/3}$ have a sharp corner at 0. For max/min functions, count the corners and breaks on the graph.

Grade - Medium

Practice map, main concept: Sheet: J-I 7; J-II 22, 52; J-M 15, 17; J-A 4; C-I 13, 15, 21; C-II 10, 23 | DPP: 4(1-3) | Question Bank: 122-124

Also used in: Sheet: J-II 30, 57-59; C-I 23; C-II 21 | DPP: 3(1, 5, 13-14); 4(12)

Quick Concept

Question 45

How to apply quick concept to check differentiability?

Model Answer

Check continuity at a first. Then LHD $= \lim_{x\to a^-} f'(x)$ (derivative of the left piece), RHD $= \lim_{x\to a^+} f'(x)$; compare.

Ex: $e^x - e + 9$ $(x < 1)$, $x^{100} + 7x^{50} + 1$ $(x \geq 1)$: continuous $(9 = 9)$, $f'(1^-) = e$, $f'(1^+) = 450$, not derivable.

Grade - High

Practice map, main concept: Sheet: J-II 42, 45, 53 | DPP: 3(1, 11, 13); 4(13)

Also used in: Sheet: J-I 13; J-II 24; J-M 7, 9, 13, 15-16, 19; J-A 3, 16 | DPP: 3(2, 7, 9, 12); 4(11) | Question Bank: 120, 122, 124, 127

Question 46

What is the condition to apply quick concept?

Model Answer

(i) f continuous at the point, (ii) each piece is easy to differentiate, (iii) $\lim f'(x)$ is not oscillatory.

Ex: not for $x^2 \sin \frac{1}{x}$ at 0: $\lim f'(x)$ oscillates, yet $f'(0) = 0$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 47
When will derivability check by quick concept be preferred over the check by basic definition?

Model Answer
When f is continuous and the pieces are standard derivable functions (polynomial, exponential, trigonometric). Use the definition for [ ], { }, oscillating terms, functions not given by formulas, or when continuity fails.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 48
How to find $f'(x)$ when $f(x)$ is a non-uniform function?

Model Answer
Differentiate each piece on its open interval; at each junction find LHD and RHD separately and include the point in f' only if they are equal.
Ex: $\frac{x^2}{4} - \frac{3x}{2} + \frac{13}{4}$ $(x < 1)$, $|x - 3|$ $(x \geq 1)$: $f' = \frac{x}{2} - \frac{3}{2}$ $(x < 1)$, -1 $(1 \leq x < 3)$, 1 $(x > 3)$; not derivable at 3.

Grade - Medium
Practice map, main concept: Sheet: J-II 56; C-I 22
Also used in: Sheet: J-II 25, 27, 29, 40, 42, 57; J-M 1; C-I 20

Question 49
How to check differentiability of a composite of two functions by defining the composite?

Model Answer
Split the domain where the inner function crosses the break points of the outer function; write $f(g(x))$ piecewise on each interval; then check continuity and LHD, RHD at the junctions.
Ex: $f(x) = |x - 1|$: $f(f(x)) = ||x - 1| - 1|$ is not derivable at $x = 0, 1, 2$.

Grade - Medium
Practice map, main concept: Sheet: J-II 25; J-M 6; J-A 3, 5, 13 | DPP: 4(10)
Also used in: Sheet: J-II 61; C-II 11, 14

Question 50
Comment on differentiability of $g(x)|f(x)|$ at $x = a$ if $f(a) = 0$.

Model Answer
(f, g derivable, a a simple zero of f.) If $g(a) = 0$: derivable at a. If $g(a) \neq 0$: not derivable (check). Same for $|f(x)|$ alone ($g = 1$).
Ex: $(x^2 - 1)|x^2 - 3x + 2|$ is not derivable only at $x = 2$ ($g(1) = 0$, $g(2) = 3$).

Grade - High
Practice map, main concept: Sheet: J-I 17; J-A 2, 6 | DPP: 3(9)
Also used in: Sheet: J-A 5, 7, 11; C-II 12

Defining of Special types of Functions

Question 51

How to define $y = \max\{f(t) : a \le t \le x\}$ or $y = \min\{f(t) : a \le t \le x\}$?

Model Answer

Sketch $f(t)$ from $t = a$. While f rises (for max) $y = f(x)$; after a peak y stays equal to that peak value until f goes above it again. Write y piecewise, then check.

Ex: $\max\{\sin t : 0 \le t \le x\} = \sin x$ for $0 \le x \le \frac{\pi}{2}$, $= 1$ for $\frac{\pi}{2} \le x \le 2\pi$; continuous at $\frac{\pi}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 16; J-M 13 | DPP: 4(12) | Question Bank: 120

Also used in: none

Determination of Function

Question 52

What are the steps to determine the function if the functional rule of a differentiable function is given?

Model Answer

(1) $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$; (2) use the rule on $f(x+h) - f(x)$ (find $f(0)$ by $x = y = 0$); (3) get $f'(x)$ in terms of $f'(0)$ and integrate; (4) fix the constant from $f(0)$.

Ex: $f(x + y) = f(x) + f(y) + 2xy - 1$, $f'(0) = 4$: $f(0) = 1$, $f'(x) = 2x + 4$, $f(x) = x^2 + 4x + 1$.

Grade - Medium

Practice map, main concept: Sheet: J-I 18-21; J-II 34-36; J-M 3, 18; J-A 12; C-I 11 | DPP: 4(4) | Question Bank: 113, 130

Also used in: none

Question 53

How to determine the function by the partial derivative approach if the functional rule of a differentiable function is given?

Model Answer

Differentiate the rule with respect to one variable keeping the other constant; then put that other variable $= 0$ (or 1) to get a relation for $f'(x)$; integrate.

Ex: $f(xy) = f(x) + f(y)$, $f'(1) = 3$: w.r.t. y: $xf'(xy) = f'(y)$; $y = 1$: $f'(x) = \frac{3}{x}$, $f(x) = 3\ln x$ $(x > 0)$.

Grade - Medium

Practice map, main concept: Sheet: J-M 4, 14 | Question Bank: 121

Also used in: none